LARGE EHV TRANSMISSION SUBSTATIONS

Design Basis, Design Basis Accident, and Military Action Extension

DBA-ES-TX-FC2: Large EHV Transmission Substation Facility Class

A Child Document Under

DBA-ES: Electric Sector Design Basis Framework (v0.5)

Version 0.1 — DRAFT

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Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. First publication of the DBA-ES-TX-FC2 facility-class instrument under DBA-ES sector parent (v0.5). Established facility class definition (345–500 kV metropolitan ring configuration; CIP-014 Tier 1 or Tier 2 designation; LPT with RT_HVLLC ≥12 months). Instantiated all four sector-level postulated event categories with FC2-specific events. Developed DBA non-adversarial consequence envelope. Applied DB→DBA→DBA-MA three-layer progression. Established RT_HVLLC as the governing consequence-duration metric for the MA extension. Incorporated Metcalf (2013) and the UK National Grid EHV substation attack (2024) as primary evidentiary anchors.

Section 1 — Purpose, Scope, and Inheritance

1.1 Purpose

This document establishes the Design Basis for large extra-high-voltage (EHV) transmission substations within the Electric Sector Design Basis Series. It is the second facility-class instrument in the series, following DBA-HY-FC1 (large hydroelectric storage facilities), and the first instrument to address TransCo's physical infrastructure directly.

The document applies the DB→DBA→DBA-MA three-layer analytical progression to EHV transmission substations serving metropolitan ring configurations: a baseline Design Basis (DB) that establishes postulated events from equipment failure, natural hazards, and operational conditions; a Design Basis Accident (DBA) layer that bounds the non-adversarial consequence envelope and establishes acceptance criteria; and a military action extension (DBA-MA) that applies the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to each postulated event category.

Large EHV transmission substations are the single facility class for which the combination of long-lead-time equipment exposure and the intentionality and precision characteristics of military action produces the most severe extended-consequence scenario in the electric sector. The large power transformer (LPT) is the governing vulnerability: a custom-manufactured component with replacement lead times of eighteen to twenty-four months under normal procurement conditions, no field repair path following physical destruction, and no strategic reserve adequate to cover simultaneous loss at multiple critical facilities. The RT_HVLLC (Recovery Timeline for High-Value Long-Lead Components) metric, established as a series standard in DBA-ES §6.2, is the primary consequence-duration metric throughout this instrument.

1.2 Facility Class Definition

A facility within the scope of DBA-ES-TX-FC2 must satisfy all three of the following criteria.

Voltage class. The substation operates at extra-high voltage, defined for this series as 345 kilovolts or above. Substations at 345 kV, 500 kV, and 765 kV are within scope. Substations operating exclusively below 345 kV are outside the scope of this instrument; they may be addressed in a subsequent facility-class instrument if their consequence profile warrants.

Metropolitan ring configuration. The substation serves as a supply node in a metropolitan ring transmission configuration — a topology in which multiple EHV substations form a ring around a major load center, collectively supplying the load pocket through a combination of radial feeds and looped circuits. The metropolitan ring configuration is the topology most susceptible to coordinated attack because the ring structure creates shared infrastructure whose failure has correlated consequences across multiple supply nodes simultaneously.

CIP-014 designation or long-lead-time equipment trigger. The facility carries a NERC CIP-014 Tier 1 or Tier 2 designation, indicating that FERC and NERC have determined that its loss or incapacitation could result in widespread instability, uncontrolled separation, or cascading failures. Alternatively, the facility satisfies Trigger 3 of DBA-ES §1.5 by containing large power transformers at EHV voltage with an RT_HVLLC of twelve months or more under normal procurement conditions. In practice, all EHV metropolitan ring substations satisfy both conditions simultaneously.

1.3 Evidentiary Anchors

Two events anchor the evidentiary foundation for FC2 and must be referenced in any analysis performed against this design basis.

Metcalf Sniper Attack (April 16, 2013). Unidentified individuals discharged rifle fire into the cooling radiators of seventeen large power transformers at PG&E's Metcalf transmission substation near San Jose, California, draining transformer oil and causing thermal shutdown. The attack lasted approximately nineteen minutes. No perpetrators were identified. The facility operated at 230 kV — below the FC2 threshold — but the attack methodology, consequence profile, and regulatory response are directly applicable to EHV facilities. The Federal Energy Regulatory Commission subsequently commissioned a classified analysis concluding that a coordinated attack on approximately nine critical transmission substations could cause blackouts affecting most of the continental United States. That analysis is the predicate for NERC CIP-014. Metcalf established that LPT external cooling systems are a viable precision-attack vector requiring minimal adversary capability and producing an RT_HVLLC-class consequence.

UK National Grid EHV Substation Attack (2024). A coordinated physical attack on a National Grid extra-high-voltage substation in England caused significant disruption to the regional electricity supply. The attack demonstrated that the Metcalf paradigm is not a one-time event specific to the United States regulatory environment, that EHV substations at the voltage class addressed by this instrument are active targets in adversary planning, and that the consequence of a successful attack at EHV voltage is qualitatively more severe than at the 230 kV Metcalf level due to the transformer replacement lead time differential.

Both events are non-adversarial-adjacent — they occurred without the four MA characteristics operating simultaneously. Neither area of denial prevented repair crews from assessing damage—neither involved precision targeting of the specific transformer element most likely to require replacement rather than repair. Neither involved sustained adversary presence to prevent restoration. The MA extension analysis in Section 5 addresses the scenario in which all four characteristics operate simultaneously against an FC2 facility.

1.4 Inherited Methodology

All sector-level methodology is inherited from DBA-ES: Electric Sector Design Basis Framework (v0.5). The following table identifies inherited elements and any FC2-specific adaptations.

Element	Source	FC2-Specific Adaptation
DB→DBA→DBA-MA three-layer progression	DBA-ES §3.1	Inherited without modification
Three required adaptations	DBA-ES §3.2	Inherited; acceptance criteria include grid topology role
Postulated event categories: PE, SE, CE, XC	DBA-ES §3.5	FC2-specific events instantiated in Section 3
Bounding event methodology	DBA-ES §4.1	Inherited; LPT is the governing long-lead-time equipment
Natural hazard compound-event requirement	DBA-ES §4.3	Seismic + wildfire-smoke as the governing compound case
Three-tier consequence taxonomy	DBA-ES §6.1	Both cascade and direct-dependency Tier 3 pathways apply
Long-lead-time equipment and an extended consequence regime	DBA-ES §6.2	RT_HVLLC is the primary FC2 consequence-duration metric
MA extension trigger criteria	DBA-ES §1.5	FC2 satisfies Triggers 1, 2, and 3 by definition
Four MA characteristics (intentionality, persistence, precision, denial)	DBA-ES §5.2	Applied to each FC2 postulated event category in Section 5
Dual Design Basis architecture	DBA-ES §3.4	Physical DB (this document) + AI Governance DB interaction acknowledged in Section 3.6

1.5 Position in Hierarchy

Document Number: DBA-ES-TX-FC2-001

Series: Electric Sector Design Basis Series — Facility Class Instruments

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.5)

Position in hierarchy: Facility-class instrument. All sector-level methodology is inherited from DBA-ES. This document applies that methodology to large EHV transmission substations and develops the RT_HVLLC consequence-duration analysis specific to this facility class.

Section 2 — Facility Profile: Large EHV Transmission Substation

2.1 Physical Configuration

A large EHV transmission substation in a metropolitan ring configuration is a substantial fixed installation: multiple transformer banks, an extensive EHV bus structure, numerous circuit breakers and disconnect switches, protective relay systems, and a control building housing the protection panels, SCADA equipment, and communications systems that govern the facility's operation. The physical footprint of a 500 kV substation typically ranges from ten to forty acres, depending on the bus configuration and the number of transformer banks.

The large power transformer is the critical element that defines the FC2 consequence profile. A 500 kV autotransformer bank is a custom-engineered component manufactured to the specific voltage ratio, MVA rating, and impedance characteristics of its application. No two installations are identical in all parameters. The custom manufacturing requirement means that a replacement cannot be taken from a shelf; it must be ordered, manufactured, factory-tested, transported on specialized heavy equipment, and installed and commissioned in the field. The full replacement timeline from order to energization is twelve to twenty-four months under normal conditions. Under supply chain stress — whether from increased demand following a coordinated attack or from adversary supply chain interdiction — the timeline extends further.

The transformer oil system is a secondary physical vulnerability. An oil-filled transformer contains thousands of gallons of mineral oil that serves simultaneously as an electrical insulator and a coolant. Oil loss from the cooling radiators — the Metcalf attack vector — removes the cooling function and causes thermal shutdown within hours. Radiators are located on the exterior of the transformer tank and are not hardened against ballistic or explosive attack in standard installations. An oil fire following radiator penetration presents both a physical damage risk to the transformer itself and a site access risk that can delay damage assessment and initiate suppression response.

2.2 Grid Topology Role

An EHV substation in a metropolitan ring configuration performs three grid functions simultaneously: it steps down EHV transmission voltage to the sub-transmission voltage used by DisCo facilities (typically 115 kV to 230 kV); it serves as a switching node in the ring topology, allowing power flows to be rerouted around contingencies; and it provides the local reactive power support that maintains voltage in the load pocket.

The metropolitan ring topology creates a shared-infrastructure vulnerability that is the defining characteristic of the FC2 class. Each substation in the ring depends on the other ring nodes to maintain load-pocket supply under contingency conditions. If one node is de-energized, the remaining nodes pick up the load through alternate ring paths. If two or more nodes are simultaneously de-energized in a coordinated attack, the ring topology's contingency coverage is exhausted, and the load pocket is deprived of supply. The intentionality characteristic of military action is precisely calibrated to exploit this topology: an adversary who understands the ring configuration can identify the minimum target set required to deprive the load pocket of supply with no switching recovery path.

2.3 Regulatory Context

NERC CIP-014 (Physical Security) is the primary mandatory regulatory instrument applicable to FC2 facilities. CIP-014 requires Transmission Owners to identify transmission stations and substations that, if rendered inoperable or damaged, could result in widespread instability, uncontrolled separation, or cascading within an interconnection. Facilities meeting this threshold are designated Tier 1 or Tier 2. They must complete a physical security risk assessment, develop and implement a physical security plan, and have that plan verified by a qualified third party.

CIP-014 addresses the physical security posture of individual facilities. It does not establish a design basis — a defined set of postulated attack conditions that the facility must be demonstrated to withstand. It does not address the RT_HVLLC consequence duration or the grid reliability posture during an extended LPT outage. It does not address the coordinated attack on multiple ring nodes simultaneously. This document supplies the design basis layer that CIP-014 does not provide.

FERC Order 693 and the associated TPL reliability standards (TPL-001 through TPL-007) establish the planning criteria that govern system performance under contingency conditions. The N-1 and N-1-1 planning standards establish what the grid must be capable of sustaining following single and double contingencies. The design basis established in this document is complementary to NERC TPL standards: TPL establishes what the system must sustain; this document establishes what the facility must withstand and how long the consequence persists when the facility fails to withstand it.

Section 3 — Design Basis: FC2 Postulated Event Categories

3.1 Sector-Level Inheritance

The four sector-level postulated event categories — PE (Power Supply and Grid Interface), SE (Structural and Physical Integrity), CE (Control and Communications), and XC (Cross-Sector Cascade) — are inherited from DBA-ES §3.5. Each is instantiated below with the specific events applicable to the FC2 facility class. No additional postulated event categories are locally defined for FC2; the four inherited categories are sufficient to bound the FC2 event space.

3.2 Category PE — Power Supply and Grid Interface Events

At FC2, power supply events are primarily events that affect the facility's transmission connections — the lines and cables that carry power to and through the substation. The facility is itself a supply node, not a load; its 'input' is the transmission system and its 'output' is the load pocket it supplies. PE events that disconnect the facility from the transmission system remove its ability to serve its load-pocket function.

PE-SS (Station Service Loss) is separately identified because it has a distinct consequence profile from other PE events. Loss of station service does not remove the substation's ability to carry power — the transformer and bus structure remain energized — but it removes the operator's ability to monitor and control the substation. The protection systems, SCADA, and communications equipment go dark. The substation continues to carry load passively but cannot respond to changing grid conditions, cannot execute switching operations, and cannot provide situational awareness to the CntlCo.

Code	Event	Initiating Condition	Governing Concern
PE-1	Single-contingency transmission loss (N-1)	Loss of a single transmission element — line, transformer bank, or bus section — connected to the substation. Protection systems clear the fault and isolate the failed element. The substation remains energized through alternate paths.	Substation must remain operational and capable of serving its load under N-1 conditions. Protection coordination must clear the fault without causing sequential tripping of additional elements.
PE-2	Double-contingency transmission loss (N-1-1)	Loss of two transmission elements, either simultaneously or within the recovery window of the first contingency. The substation may be partially de-energized or operating with reduced capacity.	Metropolitan ring substations serving as load-pocket supply nodes must demonstrate load continuity under N-1-1 through switching operations, load shedding, or load transfer. This is the NERC planning standard for high-consequence facilities.
PE-3	Complete loss of high-side bus	All transmission connections to the EHV bus are simultaneously lost, or the bus itself is faulted. The substation is de-energized at the transmission voltage level. Station service (hotel load) may	PE-3 is the bounding PE event for FC2. Loss of the EHV bus at a metropolitan ring substation removes all transmission capacity from the load pocket. Station

		remain supplied from the low-side distribution bus if it is energized from an alternate path.	service continuity determines whether protection systems, SCADA, and communications remain operational during the event.
PE-SS	Station service (hotel load) loss	Loss of AC station service power supplying control panels, relay panels, SCADA, communications equipment, and protection systems. Three sub-cases: (a) EHV bus lost, distribution low-side intact — station service maintained from low-side; (b) both high- and low-side buses lost — UPS bridges station service; (c) UPS depleted or failed — immediate control loss.	Sub-case (c) is the governing concern. Loss of station service removes the operator's ability to monitor and control the substation at the same moment that protection systems may be needed to respond to evolving fault conditions. The substation goes operationally blind under the conditions most likely to require operator action.
PE-4	Voltage excursion — high	Sustained voltage above the ANSI C84.1 Range B upper limit at the EHV bus, caused by loss of load in the load pocket (load shedding, large load trip), combined with reactive power surplus from transmission lines and capacitor banks.	Transformer insulation stress and potential failure of voltage-sensitive customer equipment. Requires automatic reactive power management response and, if not corrected, manual operator intervention.
PE-5	Voltage excursion — low	Sustained voltage below ANSI C84.1 Range A lower limit at the EHV bus, caused by heavy load demand combined with reactive power deficit. It can initiate voltage collapse in a metropolitan load pocket if not arrested by automatic or manual reactive support.	Voltage collapse is the highest-consequence PE event that does not involve physical equipment damage. It can deprive a large metropolitan area of power without damaging the substation itself, complicating restoration.

3.3 Category SE — Structural and Physical Integrity Events

Structural and physical integrity events at FC2 concentrate on the LPT as the governing consequence element. Every SE event must be analyzed for its LPT consequence because the LPT is the single element whose failure produces an RT_HVLLC-class consequence. Non-LPT SE events (circuit breaker failure, structural damage that does not involve the LPT) produce shorter-duration consequences bounded within the non-adversarial DBA. However, they must still be analyzed for their consequences in the grid topology context.

Code	Event	Initiating Condition	Governing Concern
SE-1	Large power transformer failure — internal fault	Internal fault within the LPT winding, core, or bushing, resulting in differential relay operation and transformer removal from service. The transformer may be repairable (bushing replacement, oil	RT_HVLLC governs. If the fault requires LPT replacement, the consequence duration is 18–24 months at current procurement lead times. The design basis must

		processing) or require full replacement depending on fault severity.	distinguish the repairable-fault case (days to weeks) from the replacement case (months) because the acceptance criteria differ fundamentally.
SE-2	Large power transformer failure — external physical damage	Physical damage to the LPT from fire, flooding, projectiles, or blasts. External physical damage is more likely to require full replacement than internal electrical faults. The oil-filled tank, radiators, and bushings are the most vulnerable external elements.	This is the bounding SE event for FC2 under both non-adversarial and MA conditions. External physical damage is the scenario addressed by Metcalf (2013) and CIP-014. The design basis must treat external physical LPT damage as a design basis accident, not only a beyond-design-basis event.
SE-3	High-voltage circuit breaker failure	Failure of an EHV circuit breaker to open on command during a fault-clearing sequence (breaker failure), or failure to close during a switching operation. Breaker failure initiates a backup fault-clearing sequence that trips a larger portion of the substation.	Breaker failure at an EHV metropolitan ring substation can escalate a bounded fault into a wider outage. Lead times for EHV circuit breakers are significant (6–12 months) but not at the LPT level. Protection system design must include breaker failure protection with defined backup clearing zones.
SE-4	Natural hazard — seismic	Ground motion exceeding the design basis earthquake level for the facility's geographic location. For EHV substations, seismic vulnerability concentrates in transformer bushings, disconnect switches, and rigid bus structures. Oil-filled equipment presents a fire risk following seismic damage.	Post-seismic fire at an oil-filled transformer is the compound event governing concern. Seismic damage that initiates a transformer fire under conditions where site access for suppression is impaired (road damage, aftershock hazard) extends the consequence duration beyond the physical damage alone.
SE-5	Natural hazard — extreme wind / tornado	Wind event exceeding design wind speed for transmission structures in the substation vicinity, resulting in conductor damage, structure collapse, or debris impact on substation equipment. Tornadoes present a direct strike risk to EHV bus structures and transformer installations.	A direct tornado strike on an EHV substation is a low-probability but high-consequence event that the non-adversarial design basis must bound. The consequence is similar to SE-2 (external physical damage) but the initiating mechanism is natural rather than adversarial.
SE-6	Natural hazard — compound event	Sequential or simultaneous occurrence of two natural hazard events in which recovery from the first is impaired by the second. The governing compound case for FC2 is a seismic event followed by wildfire-smoke in a Western US transmission corridor, where seismic damage to substation equipment	The compound-event case establishes the upper bound of the non-adversarial extended-consequence regime. If the compound case produces an RT_HVLLC-class consequence through natural causes, the MA extension must address how an

		coincides with aerial access restrictions and contractor mobilization delays from wildfire activity.	adversary could exploit the same vulnerability with greater certainty and longer persistence.

3.4 Category CE — Control and Communications Events

Control and communications events at FC2 do not directly de-energize the substation or damage physical equipment. Their consequences are operational: they degrade the CntlCo's ability to manage the substation and the grid in its area, remove automatic protection functions, or create conditions in which a subsequent physical event produces a larger consequence than it would have without the prior CE event. The CE events must be analyzed both in isolation and in combination with PE and SE events, because CE-induced operational degradation is the force multiplier that the MA extension's precision and denial characteristics are designed to exploit.

Code	Event	Initiating Condition	Governing Concern
CE-1	Loss of substation SCADA and remote monitoring	Loss of communications between the substation and the CntlCo energy management system, removing remote visibility into substation status (breaker positions, transformer loading, voltage and current measurements). Substation continues to operate under local protection but the CntlCo loses situational awareness.	CE-1 does not physically de-energize the substation. Its consequence is operational: the CntlCo cannot manage the transmission network in the substation's area with full situational awareness, which impairs contingency management and restoration coordination under concurrent stress conditions.
CE-2	Protective relay misoperation	Protective relay operation that does not correctly correspond to the actual fault condition: a relay that operates without a fault (false trip) or fails to operate during a fault (failure to trip). Both failure modes produce consequences at the substation and potentially at the grid level.	False trips at an EHV metropolitan ring substation can initiate load shedding and voltage instability without a physical fault. Failure to trip allows a fault to persist until backup clearing, which may involve tripping a larger zone. Both are design basis events that protection system design must demonstrate to be within acceptable probability bounds.
CE-3	Energy management system (EMS) failure at CntlCo	Loss or severe degradation of the CntlCo's energy management system, including state estimator, contingency analysis, and automatic generation control functions. The CntlCo operates in a degraded situational awareness mode that increases the risk of undetected constraint violations.	CE-3 is a CntlCo event with FC2 consequences: the substation's operating state is unchanged, but the grid-level management capability that interprets that state and coordinates responses to contingencies is degraded. Under concurrent physical stress at FC2

			facilities, CE-3 compounds the consequence.
CE-4	Cyber intrusion into substation control systems	Unauthorized access to substation automation systems, relay settings, or SCADA remote terminal units. The intrusion may be passive (intelligence collection, pre-positioning) or active (relay setting modification, breaker state manipulation). CIP-013 supply chain risk is the primary entry pathway.	Active cyber intrusion that modifies relay settings is analytically equivalent to CE-2 (protective relay misoperation) with adversarial intent. The design basis must address both the non-adversarial misoperation case (CE-2) and the intentional manipulation case (CE-4 in the MA extension bridge).

3.5 Category XC — Cross-Sector Cascade Events

Cross-sector cascade events at FC2 operate through two distinct pathways established in DBA-ES §6.1: the cascade pathway (grid-level degradation propagating into dependent sectors) and the direct-dependency pathway (supply disruption reaching a critical function without grid-level cascade). Both pathways are present and analytically significant for FC2.

The cascade pathway (XC-1, XC-2) is the scenario addressed by the NERC CIP-014 consequence logic: the facility's loss produces widespread instability or cascading. The direct-dependency pathway (XC-3) is the scenario that CIP-014 does not address: the facility's loss removes supply from a defense-critical load through direct connection, without requiring a grid cascade. Both pathways produce Tier 3 consequences and both are within the FC2 design basis.

Code	Event	Initiating Condition	Governing Concern
XC-1	Metropolitan load pocket deprivation	Loss of the FC2 substation removes primary transmission supply to a metropolitan load pocket. Cascading consequences include activation of demand response resources, voltage depression across the load pocket, loss of supply to DisCo substations served from the facility, and potential motor stall in the load pocket if voltage depression is sustained.	The cascade pathway to Tier 3 runs through the DisCo substations that receive power from the FC2 facility. Water treatment, hospital facilities, emergency communications, and defense-critical loads served by those DisCo substations are the ultimate cross-sector consequence targets.
XC-2	Grid cascade from FC2 loss in a constrained corridor	Loss of an FC2 facility in a transmission corridor that is already operating near thermal or stability limits triggers automatic protection actions (line tripping, load shedding) that propagate across the interconnection. The cascade is not initiated by the FC2 loss alone but by the FC2 loss under conditions of a	This is the scenario that produced the 2003 Northeast Blackout: the initiating event was modest, but the pre-existing constraint conditions transformed it into a cascade. The FC2 design basis must establish the conditions under which the facility's loss

		prior constraint that the grid operator was managing.	transitions from a local contingency to a cascade initiator.
XC-3	Defense-critical load disruption via direct supply	Loss of the FC2 facility directly removes EHV supply to one or more loads that are military installations, continuity-of-government sites, or defense-critical manufacturing facilities. The Tier 3 consequence is reached through direct supply interruption, not grid cascade. Trigger 4 of DBA-ES §1.5 fires on this condition.	XC-3 is the direct-dependency Tier 3 pathway for FC2. Its consequence analysis must identify the specific defense-critical loads within the substation's service area, their backup power posture and duration, and the condition at which backup posture is exhausted and an unrecoverable mission consequence begins.
XC-4	Natural gas generation loss from the gas-electric cascade	Loss of an FC2 facility serving gas-fired GenCo generation in its supply area reduces electric supply for compressor station auxiliaries. Reduced compressor availability reduces gas pressure in supply pipelines. Reduced gas supply curtails gas-fired generation output. The cascade is self-reinforcing once initiated.	The gas-electric cascade is the XC event that most clearly demonstrates the bidirectional dependency structure of the electric sector. The FC2 design basis must identify gas-fired generation assets in the substation's service area and the pipeline infrastructure that those assets depend on, so the cascade pathway can be modeled.

3.6 Dual Design Basis Architecture

Consistent with DBA-ES §3.4, an FC2 facility may simultaneously carry a sector-specific physical Design Basis (this document) and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level for FC2 in three specific operational contexts.

EMS and state estimation. CntlCo energy management systems increasingly incorporate ML-based state estimation and contingency analysis tools. These tools operate below the Brightline — they are deterministic, testable ML systems — and may operate autonomously. However, their outputs govern CntlCo decisions about FC2 facility dispatch and contingency management. A postulated physical event at an FC2 facility that simultaneously degrades the ML state estimator's input data (through SCADA loss, CE-1) removes the CntlCo's analytical basis for response decisions at the same moment that response is most needed.

Protection relay setting optimization. ML tools are beginning to be applied to protective relay setting optimization for complex transmission topologies. These tools are ML (not Gen AI) and operate below the Brightline. However, if relay settings produced by ML optimization are incorrect or have been adversarially manipulated (CE-4 in the MA extension), the error may not be discoverable through standard testing because the ML tool and the relay system share the same data environment.

Operational AI under degraded conditions. Any Gen AI application supporting FC2 operational decision-making requires a human Command Broker above the Brightline per DBA-ES §3.4. Under the denial characteristic of military action, the Command Broker may be unreachable (communications disruption) or operating under duress. The design basis must establish minimum human oversight requirements that remain enforceable under degraded communications conditions.

Section 4 — Design Basis Accidents: Non-Adversarial Envelope

4.1 Bounding Event Methodology

Bounding events for FC2 are identified through the methodology established in DBA-ES §4.1: physical plausibility, historical frequency, and consequence severity. Within each postulated event category, the governing case is the event configuration that produces the highest consequence at each tier of the consequence taxonomy.

For FC2, the governing case within SE is unambiguous: SE-2 (LPT external physical damage requiring replacement) produces the highest consequence at every tier. It produces Tier 1 consequences (facility de-energized), Tier 2 consequences (metropolitan load pocket loss), and Tier 3 consequences (cross-sector cascade or direct dependency disruption) simultaneously, and it does so on an RT_HVLLC-class duration. SE-2 is therefore the design basis accident for FC2.

4.2 Non-Adversarial Consequence Envelope

The non-adversarial design basis accident envelope for FC2 is bounded by the following conditions, organized by consequence tier.

Tier 1 — Facility-level consequences. The bounding Tier 1 consequence is loss of all EHV transformer capacity at the facility for a duration equal to RT_HVLLC (18–24 months) following LPT replacement-class damage. Sub-transmission capacity may remain partially available if the low-voltage bus structure is undamaged. Station service continuity is maintained through UPS for the UPS design duration; beyond that duration, station service is lost unless a dedicated station service supply remains available.

Tier 2 — Grid-level consequences. The bounding Tier 2 consequence is metropolitan load pocket deprivation for a duration governed by the availability of alternate supply paths through the ring topology. Under N-1-1 conditions at the metropolitan ring level, the load pocket may be partially supplied through alternate paths at reduced capacity with voltage depression and load curtailment. Complete load pocket deprivation occurs when the FC2 loss eliminates all ring supply paths simultaneously.

Tier 3 — Cross-sector consequences. The bounding Tier 3 consequence is exhaustion of backup power posture at the most time-constrained dependent critical facility in the substation's service area. For defense-critical facilities with a generator backup of 72 hours of fuel, the Tier 3 acceptance criterion requires either grid restoration or fuel resupply within 72 hours of FC2 loss. For facilities with UPS-only backup of minutes to hours, the acceptance criterion is more stringent. It may not be satisfiable by any recovery action if the FC2 loss is not reversed quickly.

4.3 Extended Consequence Regime

The non-adversarial DBA for FC2 must formally address the extended consequence regime defined in DBA-ES §6.2. An LPT replacement-class event at FC2 produces a consequence that persists on the RT_HVLLC timeline regardless of whether the initiating event was adversarial. The extended consequence regime analysis must establish the grid reliability posture during the RT_HVLLC period: which compensating measures are available, what transmission constraints must be managed, and which cross-sector dependency thresholds are at risk within the restoration window.

The critical question in the extended consequence regime is whether the interconnection can sustain the FC2 outage without cascading consequences over the RT_HVLLC period. This question is not asked by NERC planning standards, which address the short-term contingency response rather than the multi-month sustained outage condition. It is a design basis question that this instrument requires to be answered for each FC2 facility as part of the site-specific implementation.

Section 5 — Military Action Extension

5.1 MA Extension Applicability

All FC2 facilities satisfy at least three of the six MA extension trigger criteria established in DBA-ES §1.5 by definition. Trigger 1 fires because the governing failure mode produces Tier 2 or Tier 3 consequences. Trigger 2 fires because CIP-014 Tier 1 or Tier 2 designation is a facility class criterion. Trigger 3 fires because EHV LPTs have RT_HVLLC ≥12 months. FC2 facilities in regions with elevated threat environments also satisfy Trigger 6, and those serving defense-critical loads directly satisfy Trigger 4. The MA extension is mandatory for all FC2 facilities. No FC2 facility is DBA-complete without the MA extension.

5.2 Application of the Four Characteristics

The four MA characteristics — intentionality, persistence, precision, and denial — are applied to each postulated event category in the following governing scenarios table. The table identifies how each characteristic modifies the non-adversarial design basis event into a military action scenario and what the resulting consequence-duration implication is.

Category	Governing Military Action Scenario	Application of Four Characteristics	RT_HVLLC Implication	Acceptance Criterion
PE	Simultaneous multi-element transmission attack targeting the metropolitan ring configuration to isolate the load pocket	Intentionality: adversary selects the N-k combination that produces load pocket isolation, not random element loss. Persistence: Access denial prevents switching operations and emergency restoration. Precision: target selection optimizes for maximum load impact. Denial: road closures and area denial prevent operator access to the substation for manual switching.	Not applicable to PE events directly. RT_HVLLC governs if PE event damages LPT (see SE).	Load pocket isolation must not produce unrecoverable cross-sector consequences within the backup posture duration of dependent critical facilities.
SE	Precision kinetic attack on LPT at CIP-014 Tier 1 facility — Metcalf attack paradigm scaled to EHV voltage class	Intentionality: adversary selects LPT as the highest-RT_HVLLC element. Persistence: 18–24 months replacement lead time is the adversary's effective persistence without further action. Precision: LPT external cooling radiators and bushings are the specific vulnerable elements that disable the transformer without destroying it. Denial: security perimeter degradation from an attack prevents rapid damage assessment and repair crew access.	RT_HVLLC = 18–24 months for a destroyed 500 kV autotransformer bank. This is the governing consequence-duration metric for FC2 MA extension. No NERC standard or CIP requirement addresses the grid reliability posture over an 18–24 month FC2 outage.	Physical hardening of LPT must reduce RT_HVLLC below the duration at which cross-sector Tier 3 consequences become unrecoverable for the highest-dependency loads in the substation's service area.
CE	Coordinated cyber-physical attack: relay setting modification combined with physical damage to communications backhaul	Intentionality: adversary studies protection coordination architecture to identify relay setting changes that cause maximum consequence without an obvious signature. Persistence: modified relay settings persist until discovered through testing. Precision: cyber intrusion modifies specific relay parameters to cause a specific failure mode on demand. Denial: communications backhaul damage removes SCADA visibility simultaneously, preventing remote detection of relay setting changes.	CE MA scenario does not directly engage RT_HVLLC but removes the operational infrastructure needed to manage the grid during an RT_HVLLC-class SE event.	Relay settings must be verifiable through an independent out-of-band channel that the same intrusion vector cannot compromise as the primary control system.
XC	Coordinated attack on FC2 facility and DisCo	Intentionality: adversary targets both the bulk supply node (FC2) and the distribution last-mile	RT_HVLLC at FC2 governs the upper bound of consequence	FC2 design basis must be coordinated

	substations serving defense-critical loads in the same service area	simultaneously to prevent the cascade from being bridged by utility switching operations. Persistence: RT_HVLLC at FC2 sustains the load pocket isolation; DisCo outage prevents any restoration path below the transmission tier. Precision: target set is optimized to maximize the overlap between transmission outage area and defense-critical load locations. Denial: Area denial prevents utility repair crews from reaching either the FC2 facility or the DisCo substations.	duration. DisCo substation recovery timeline governs the lower bound — if DisCo can restore before FC2, partial recovery is achievable.	with FC5 design basis for the DisCo substations in the same service area, so that the coordinated-attack scenario is covered analytically by the two instruments together.

5.3 The Metcalf Paradigm Scaled to EHV

The Metcalf attack (2013) demonstrated that precision physical attack on LPT cooling systems is a viable attack vector producing an RT_HVLLC-class consequence at sub-EHV voltage. At EHV voltage, the consequence is qualitatively more severe for three reasons.

First, EHV autotransformers are larger, more complex, and more difficult to replace than the distribution-class and sub-transmission-class transformers at Metcalf-voltage facilities. The RT_HVLLC for a 500 kV autotransformer bank is longer than for a 230 kV unit, and the population of manufacturers capable of producing a replacement is smaller.

Second, EHV metropolitan ring substations serve larger load pockets than sub-transmission substations. The load pocket consequence of an EHV metropolitan ring substation loss is proportionally larger.

Third, EHV substations are CIP-014-designated facilities with physical security plans, whereas many sub-transmission substations are not. An adversary choosing between a secured CIP-014 facility and an unsecured sub-transmission facility with equivalent load consequence may prefer the unsecured target — but a coordinated attack across multiple substations in a metropolitan ring includes both the hardened EHV nodes and the unsecured sub-transmission nodes in the ring. The hardening of the EHV node alone does not protect the ring topology if sub-transmission nodes in the ring are vulnerable.

5.4 Coordinated Multi-Node Attack

The MA extension must address the coordinated attack on multiple nodes in the metropolitan ring simultaneously. A single-node attack at an FC2 facility produces a consequence that the ring topology can partially absorb through switching. A coordinated attack on two or more ring nodes simultaneously exhausts the ring topology's contingency coverage and produces load pocket deprivation with no switching recovery path.

The coordinated multi-node attack is not a beyond-design-basis event for FC2. It is the governing MA scenario precisely because intentionality means the adversary will identify and target the minimum node set required to defeat the ring topology. The design basis must address this scenario and establish what hardening measures reduce the probability of a simultaneous successful attack across the minimum target set to an acceptable level.

Section 6 — Consequence Analysis

6.1 Consequence Chain: SE-2 Governing Case

The consequence chain for the governing design basis accident (SE-2: LPT external physical damage requiring replacement) proceeds as follows.

Initiating event. LPT cooling radiators are penetrated by ballistic, explosive, or other physical attack. Transformer oil drains from the cooling circuit. Oil loss is detected by the Buchholz relay or sudden pressure relay, which initiates differential protection tripping to de-energize the transformer.

Immediate facility consequence (Tier 1). The transformer is removed from service. If no alternative transformer capacity is available at the facility, the high-side EHV bus serving the transformer's load is de-energized. Station service continuity depends on whether an alternate low-side supply path exists. If station service is lost, the facility goes operationally blind.

Grid consequence (Tier 2). The metropolitan ring topology redistributes load from the lost transformer to adjacent substations. If the redistribution is within the ring's N-1-1 contingency capacity, the load pocket remains partially supplied with potential for voltage depression and load curtailment. If the redistribution exceeds the ring's contingency capacity, load is shed in the load pocket.

Cross-sector consequence (Tier 3). Load shedding in the load pocket removes power from DisCo distribution substations, which in turn removes power from end-use customers, including critical facilities. Critical facilities transition to backup power. The Tier 3 consequence escalates as backup power duration is exhausted: hospitals extend generator runtime, water treatment switches to backup pumping, and defense-critical facilities activate emergency power protocols. The Tier 3 acceptance criterion is the maximum duration before any critical facility's backup posture is exhausted and the mission consequence becomes unrecoverable.

Extended consequence regime. LPT replacement is initiated. Under normal procurement, the replacement timeline (RT_HVLLC) is eighteen to twenty-four months. During this period, the load pocket operates at reduced transmission capacity with ongoing constraint management by the CntlCo. The grid reliability posture during the RT_HVLLC period must be explicitly analyzed for each FC2 facility as part of the site-specific design basis implementation.

6.2 MA Extension Consequence Chain

The MA extension consequence chain for FC2 differs from the non-adversarial chain in four ways corresponding to the four MA characteristics.

Intentionality extends the consequence from a single-node event to a multi-node event. The adversary targets the minimum node set in the metropolitan ring required to defeat load-pocket contingency coverage. The consequence is load pocket deprivation rather than partial supply reduction.

Persistence extends the consequence duration beyond the natural recovery timeline. The adversary denies access for damage assessment and replacement equipment delivery, activates supply chain interdiction to extend LPT procurement lead times, and targets replacement equipment before installation. The effective RT_HVLLC under adversarial persistence is longer than the commercial baseline.

Precision reduces the adversary capability requirement and increases the probability of success. The attack is targeted at the specific LPT element most likely to require replacement rather than repair (the bushing, the main tank weld seams, the radiator bank connections). This reduces the attack footprint while maximizing consequence duration.

Denial compounds all other consequence factors. Denial of site access prevents damage assessment, prevents repair crew operations, and prevents replacement equipment installation. Denial of communications removes CntlCo situational awareness. Denial of supply chain access extends RT_HVLLC. The MA extension consequence chain for FC2 under all four characteristics simultaneously is qualitatively more severe than any non-adversarial event in the DBA.

Section 7 — Acceptance Criteria

7.1 Tier 1 Acceptance Criteria

The facility must be capable of maintaining station service continuity for a minimum of seventy-two hours following loss of all EHV and sub-transmission connections, through UPS or a dedicated station service supply that is independent of the main transformer banks. Station service continuity enables protection system operation, SCADA function, and communications with the CntlCo during the initial emergency period.

The facility’s protection systems must operate correctly under N-1 and N-1-1 contingency conditions. Protective relay settings must be verified to produce correct responses for all fault types and locations within the protection zone. Relay settings must be verifiable through an out-of-band channel independent of the primary control system.

7.2 Tier 2 Acceptance Criteria

The metropolitan ring configuration must maintain load-pocket supply under N-1-1 contingency conditions, with load curtailment not exceeding the NERC TPL planning criteria thresholds. A single FC2 facility loss must not, under any pre-contingency operating condition within the TPL planning envelope, produce uncontrolled separation or cascading in the interconnection.

The CntlCo must be capable of managing the grid in the FC2 facility's area during the first seventy-two hours following FC2 loss, including executing load shedding and switching operations required to maintain interconnection stability. This requires that CntlCo communications with adjacent facilities remain functional during the FC2 loss event.

7.3 Tier 3 Acceptance Criteria

The maximum duration of power interruption to any defense-critical facility in the FC2 substation's service area must not exceed that facility's backup power posture duration before mission-unrecoverable consequences occur. This criterion requires that the service-area critical load inventory (DI-4) be completed by the facility operator so that the most time-constrained critical load governs the Tier 3 acceptance criterion.

If the most time-constrained critical load in the service area has a backup posture of less than seventy-two hours, the FC2 facility cannot satisfy the Tier 3 criterion through grid restoration alone within that window under RT_HVLLC conditions. In this case, the Tier 3 criterion must be satisfied through a combination of grid restoration for short-duration contingencies and alternative supply arrangements (mobile generation, dedicated feeder from alternate substation) for RT_HVLLC-class events.

7.4 Extended Consequence Regime Acceptance Criteria

The grid reliability posture during the RT_HVLLC period must be analyzed and documented for each FC2 facility. The analysis must identify which transmission constraints are binding during the outage, what compensating measures are available and at what cost, and which cross-sector dependency thresholds are at risk within the RT_HVLLC window.

This analysis is a site-specific design basis implementation requirement. It cannot be performed at the series level because the grid topology, load composition, and critical dependency profile of each FC2 facility are facility-specific. The guidance framework for this analysis will be developed in a companion implementation guide.

Section 8 — Deferred Items

Item	Description	Resolution Criterion
DI-1	Quantitative RT_HVLLC calibration. This document establishes RT_HVLLC at 18–24 months for a 500 kV autotransformer bank based on industry-reported lead times. The range should be calibrated against current procurement data from major transformer manufacturers and from the DOE/NERC STEP program inventory.	Updated RT_HVLLC range supported by manufacturer lead time data and STEP inventory analysis incorporated at next revision.
DI-2	STEP program coverage gap analysis. The DOE/NERC Spare Transformer Equipment Program provides a limited inventory of pre-positioned spares. This document asserts that STEP coverage does not address all critical EHV transformer configurations, consistent with DBA-ES §6.2. A formal gap analysis comparing FC2 facility transformer specifications against STEP inventory is needed to bound the population of FC2 facilities without a STEP-compatible spare.	STEP gap analysis completed; FC2 facilities without STEP coverage identified and logged as a highest-priority hardening recommendation.
DI-3	Space weather / GIC postulated event. Geomagnetically induced currents (GIC) from severe space weather (Carrington-class event) are a postulated event category for EHV transformers that is not yet addressed in this document. GIC causes transformer saturation, reactive power absorption, and potential thermal damage to LPT windings. The GIC postulated event is identified here as a deferred item for the next revision.	GIC / space weather postulated event category added to SE events in next revision, with reference to NERC GMD standards and applicable research literature.
DI-4	Service area defense-critical load inventory. Section 4.5 (XC-3) requires identification of defense-critical loads within the FC2 substation's service area and their backup power posture. This identification requires coordination with facility operators and, for sensitive military installations, with appropriate government agencies. It cannot be completed at the series level and must be performed by the facility operator for each FC2 facility.	Service-area critical load inventory completed by each FC2 facility operator as part of their site-specific design basis implementation. Guidance on the inventory methodology to be developed in a companion implementation guide.
DI-5	Dual-DB AI Governance interaction points. Section 3.6 acknowledges the dual-DB architecture per DBA-ES §3.4. The specific AI governance interaction points for FC2 — ML systems used in EMS state estimation, protection relay setting optimization, and load forecasting — are identified but not yet fully analyzed. The Brightline boundary between ML (autonomous) and Gen AI (command-broker-required) applications must be drawn explicitly for FC2 operational AI contexts.	AI governance interaction points for FC2 EMS and protection system AI applications identified and mapped to Brightline / Command Broker architecture at next revision.

Section 9 — References

Regulatory and Standards References

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